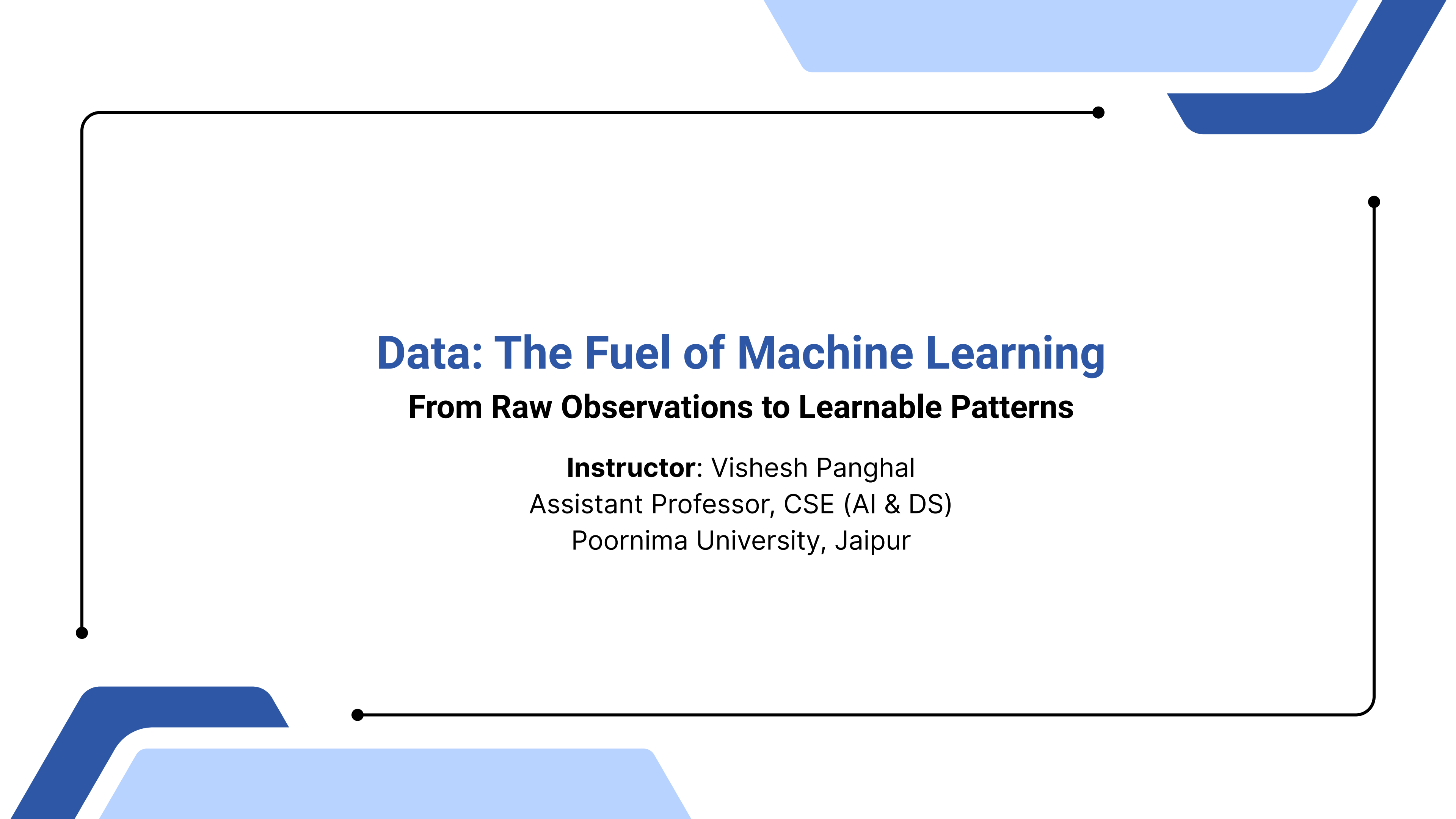




Data: The Fuel of Machine Learning

From Raw Observations to Learnable Patterns

Instructor: Vishesh Panghal
Assistant Professor, CSE (AI & DS)
Poornima University, Jaipur

In the previous lecture, we studied:



Intelligence



Human vs Machine Learning



Why Traditional Programming Fails



What is Machine Learning?



Types of Machine Learning

If machines learn from data, what exactly is data?



Data = Information = Knowledge?

DATA

- Raw Facts
- No Context
- Unprocessed

Example:

40 30 37 42



Information

- Meaningful Data
- Organized Context
- Easy to Understand

Day	Temp (°C)
Mon	40
Tue	30
Wed	37
Thu	42



Knowledge

- Actionable Insight
- Decision Making
- Based on Experience

- Heat wave trend detected.
- Take preventive measures against heat-related illnesses.

Dataset

A dataset is a collection of examples used to train and evaluate machine learning models.

The diagram illustrates a dataset table with the following structure:

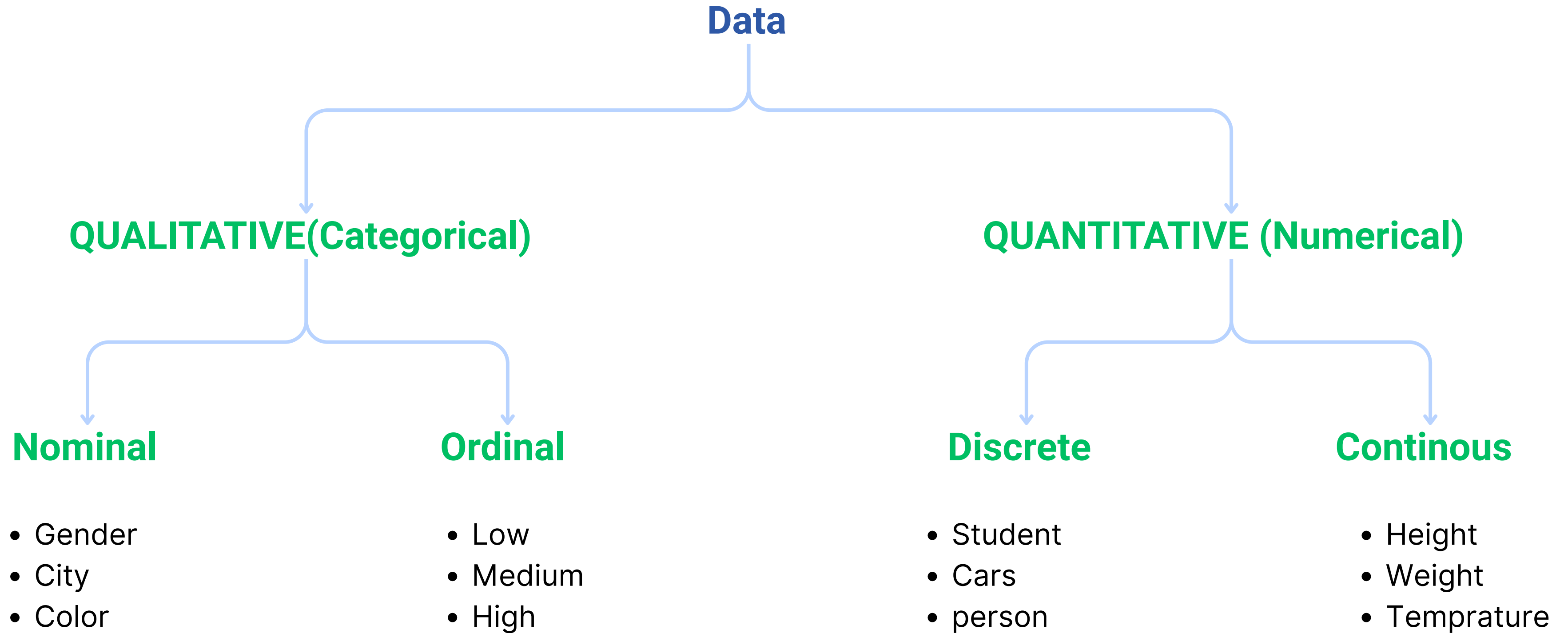
	Name	Age	Marks	Result
Rows (Records / Instances)	Vanshika	20	85	Pass
	Rahul	19	53	Fail
	Sumit	21	46	Fail
	Vinit	20	78	Pass
	Neha	19	49	Fail

Annotations:

- Columns (Attributes):** Points to the header row (Name, Age, Marks, Result).
- Rows (Records / Instances):** Points to the data rows (Vanshika, Rahul, Sumit, Vinit, Neha).
- Features (Input Variables):** Points to the columns Name, Age, and Marks.
- Label (Target Variable):** Points to the Result column.

Which column are we trying to predict?

Type of Data



Type of Data by Structure

Structured

- Fixed Schema
- Rows & Columns
- Easy to Query

Example:

Day	Temp (°C)
Mon	40
Tue	30
Wed	37
Thu	42

- SQL Tables
- Excel Sheets
- CSV Files

Semi - Structured

- Partial Structure
- Tags / Key-Value
- Flexible Format

Example:

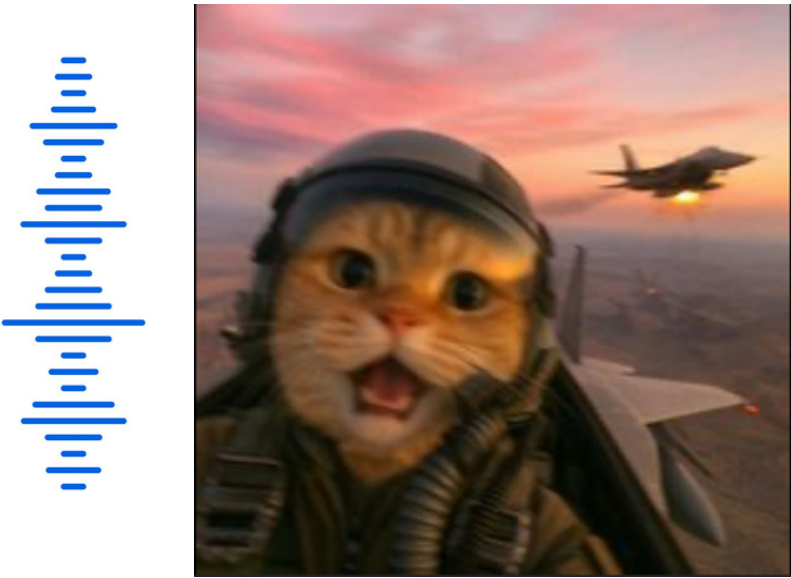
```
1 {
2   "name": "Vishesh",
3   "age": 23,
4   "is_active": true,
5   "skills": ["Computer vision", "Python", "Flutter"],
6   "address": {
7     "State": "Haryana",
8     "country": "India"
9   }
10 }
```

- JSON Files
- XML Files
- HTML Files
- Log Files

Unstructured

- No Fixed Structure
- Free-form Content
- Difficult to Analyze

Example:



- Text Documents
- Images
- Videos
- Audio

DATASET QUALITY ISSUES

Missing Values

Duplicate Data

Invalid Values

Noisy Data

Outliers

Class Imbalance

BALANCED DATASET

- Pass : 50%
- Fail : 50%



Equal class distribution

Better learning

IMBALANCED DATASET

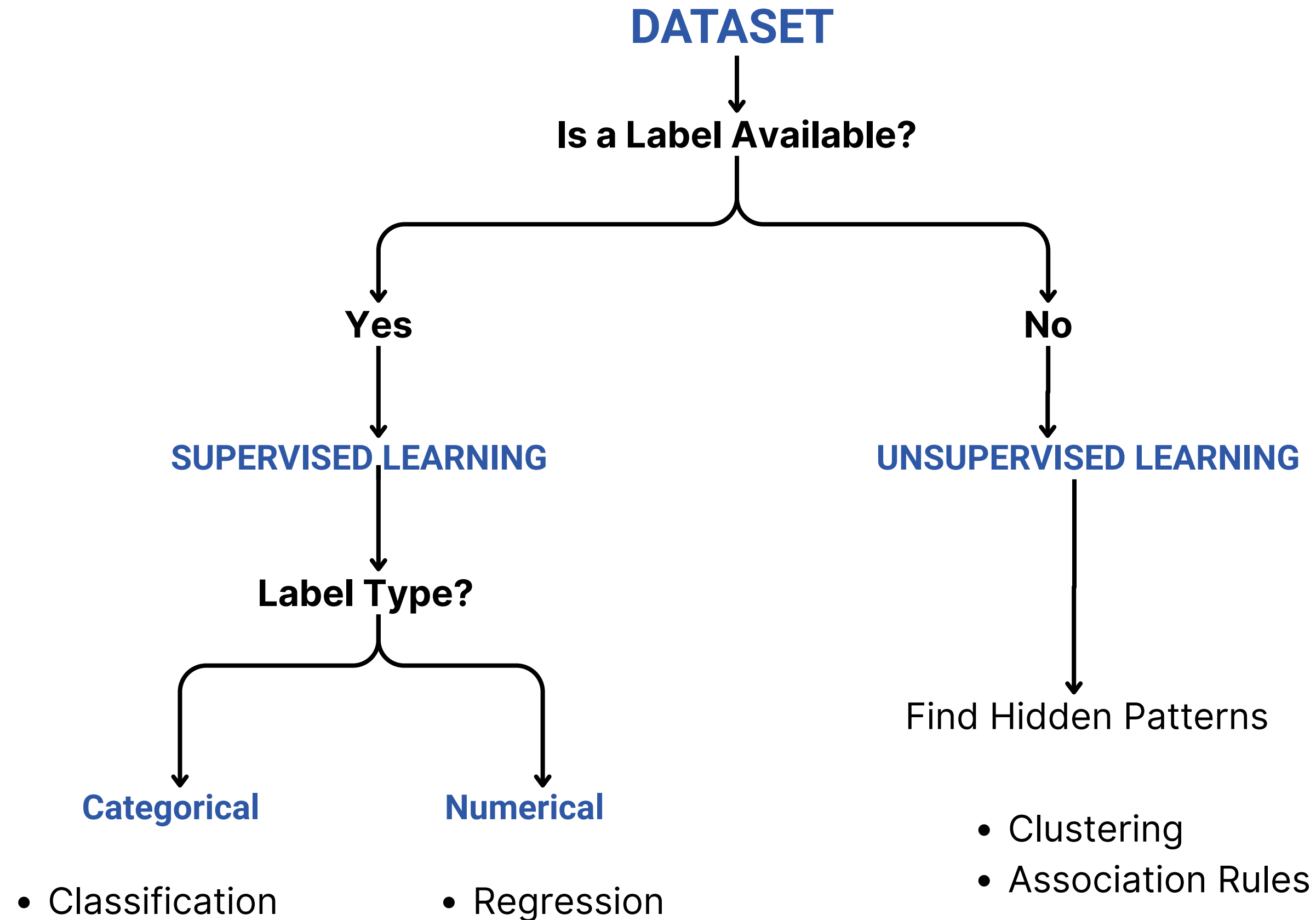
- Pass : 95%
- Fail : 5%



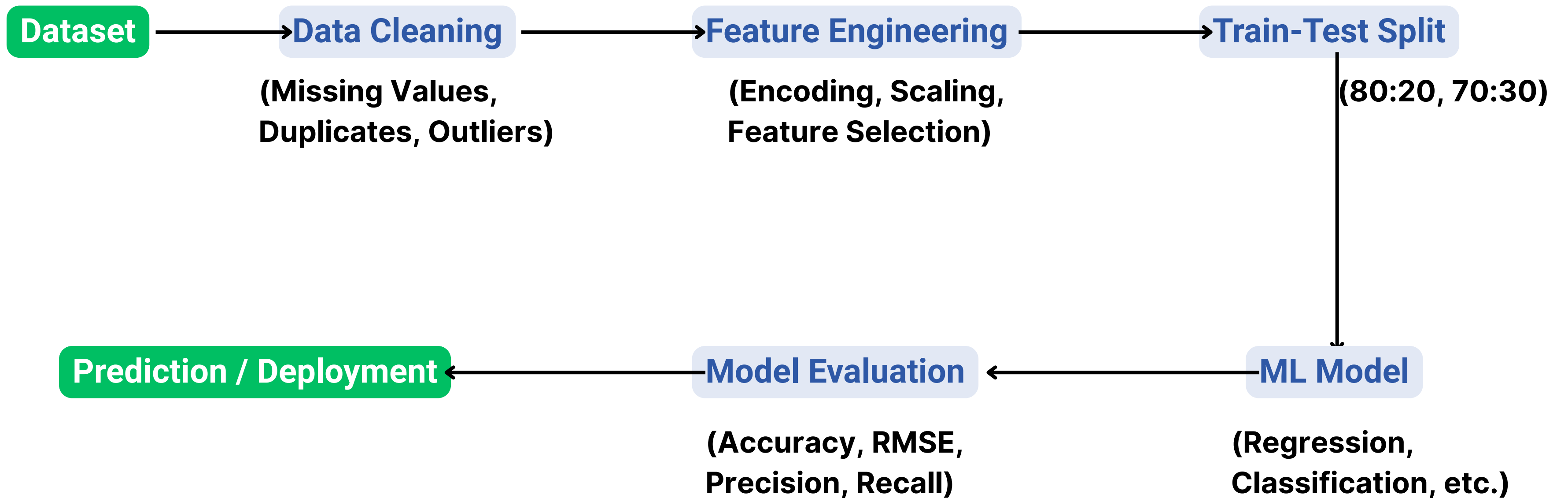
Unequal class distribution

Biased predictions

How Data Decides ML Type



MACHINE LEARNING PIPELINE





Thank You